# JEEVAN SAI REDDY CHOPPA

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## Summary

MS in Data Science graduate (University of Maryland, May 2026) with three end-to-end Large Language Model (LLM) and machine learning systems: a Retrieval-Augmented Generation (RAG) pipeline achieving 0.98 context recall with 3× lower search latency, QLoRA fine-tuning of Llama-3-8B on a self-curated 50K-example corpus, and an agentic AutoML engine validated across 5 datasets with zero code changes.

## Technical Skills

**Languages & Data:** Python, SQL, pandas, NumPy, PySpark, PostgreSQL, ETL & data pipelines
**Machine Learning:** PyTorch, scikit-learn, XGBoost, LightGBM, feature engineering, model evaluation & validation, statistical analysis
**LLMs & NLP:** Retrieval-Augmented Generation (RAG), fine-tuning (LoRA, QLoRA, PEFT), Hugging Face Transformers, LangChain, LlamaIndex, AI agents & tool calling, prompt engineering, embeddings & vector databases (Weaviate), LLM evaluation (RAGAS), spaCy, BERT
**Cloud & Tools:** AWS, Docker, Git, Ollama, vLLM, Streamlit

## Projects

**OmniData: Agentic AutoML & Conversational Analytics Engine** | GitHub *Python, LangChain, XGBoost*

- Built an agentic AutoML system that takes any raw CSV and runs end to end with zero manual code changes: LLM-based dataset profiling, automated preprocessing, and a model tournament across 10+ algorithms (XGBoost, LightGBM, linear baselines).
- Prevented data leakage, overfitting, and underfitting via three guard gates in global preprocessing (including a two-phase correlation guard) and a Pydantic-validated schema layer that adapts the pipeline to unseen datasets.
- Implemented LLM-driven diagnostics and reporting: LLaVA (multimodal) inspects OLS residual plots for non-linearity and Llama 3 interprets Macro-F1 deltas to generate the final model summary report.
- Sandboxed all LLM-generated code with a self-correction loop (up to 3 retries) and a critic validator node that checks the winning model for leakage, overfitting, and underfitting, surfacing its findings in the user-facing summary.
- Tested across 5 datasets (synthetic and real, spanning classification and regression) with zero code changes; the preprocessing gates held in every run and the critic validator never flagged leakage or overfitting.

**DeepRAG: Retrieval-Augmented Generation (RAG) Pipeline** | GitHub *Python, Weaviate, Docker, RAGAS*

- Built a Retrieval-Augmented Generation (RAG) pipeline for 250+ page research PDFs, using LlamaParse for document parsing and semantic chunking (via Gemma-3) in place of fixed token splitting.
- Cut vector storage and search latency ∼3× versus the full 768-dimension baseline by truncating nomic-embed-text embeddings to 256 dimensions with Matryoshka Representation Learning.
- Implemented hybrid search (dense embeddings + BM25) in a Dockerized Weaviate vector database, augmented with multi-query expansion and cross-encoder reranking.
- Evaluated the pipeline with RAGAS on a held-out question set, measuring 0.98 context recall and 0.93 context precision, with DeepSeek-R1 as the generation model.

**DataAdapt: Fine-Tuned Data Science Code Assistant** | GitHub *Python, Hugging Face PEFT, QLoRA, Ollama*

- Fine-tuned Llama-3-8B into a domain-specific data science code assistant using supervised fine-tuning (SFT); built the full training-data pipeline, curating, cleaning, and ChatML-formatting a 50K-example instruction corpus.
- Built a QLoRA training pipeline (Hugging Face PEFT, 4-bit quantization) that cut GPU memory use by ∼75%, making full training runs feasible on a single consumer GPU.
- Built a held-out evaluation harness benchmarking fine-tuned outputs against base Llama-3-8B on SQL and pandas code-generation tasks, scoring with ROUGE-L and BLEU.
- Reduced catastrophic forgetting by mixing general-instruction regularization data into the training set, preserving base-model performance on non-domain tasks.

## Education

**University of Maryland, College Park** College Park, MD
*Master of Science in Data Science | GPA: 3.87/4.0* *Aug 2024 – May 2026*

*Coursework:* Machine Learning, Natural Language Processing, Deep Learning, Big Data Systems, Cloud Computing

**Sree Vidyanikethan Engineering College** Tirupati, India
*Bachelor of Technology, Electronics & Communication Engineering | GPA: 3.67/4.0* *May 2024*